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PHILOSOPHY OF ARTI- FICIAL INTELLIGENCE

UNFINISHED LECTURE NOTES

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Accountability

2026-07-20 · jolly date Storch

THE RESPONSIBILITY GAP

A VISITING TRAVELLER witnesses a penal machine execute a condemned man for a sentence no living person can read or explain, and is invited to endorse the system.

ALGORITHMIC GOVERNANCE and black-box AI in high-stakes domains highlight opacity and accountability gaps: audits, explainability requirements, and governance mechanisms all attempt to close the distance between automated decision and human answerability.

Cases: when no one is answerable. Kafka's machine is a fable; the responsibility gap is not. Each system below caused real harm, and in each the question "who is accountable?" had no clean answer.

- **Amazon's hiring engine (2018).**¹ A model trained on a decade of engineering hires learned to downgrade résumés containing the word "women's" and to reward male-coded phrasing. No one wrote the rule; the bias was inherited from the data. Amazon quietly scrapped the tool. Who answers for discrimination nobody authored?
- **Google Photos (2015).**² An image classifier labelled photographs of Black users as "gorillas". The remedy was not a corrected model but a deleted label: "gorilla" was removed from the product. The error was hidden, not fixed.
- **Tesla Autopilot (2016).**³ A driver was killed when the system failed to distinguish a crossing trailer from bright sky and did not brake. Liability fractured three ways: the driver who trusted a feature named "Autopilot", the manufacturer that named it so, and the regulator that permitted the deployment.
- **The Flash Crash (2010).** Automated trading erased roughly a trillion dollars in minutes; no human decided to sell at scale and

IN DER STRAFKOLONIE
Franz Kafka (1919)



MINORITY REPORT A pre-crime unit arrests people before they offend, based on algorithmic prediction. When the system targets its own chief, the question of who holds the algorithm accountable becomes impossible to avoid.

S. Spielberg. *Minority report*. Film, 2002. 20th Century Fox; based on the short story by Philip K. Dick (1956)

¹ J. Dastin. Amazon scraps secret AI recruiting tool that showed bias against women. Reuters, 10 October 2018, 2018. [reuters.com](https://www.reuters.com)

² T. Simonite. When it comes to gorillas, google photos remains blind. Wired, 11 January 2018, 2018. [wired.com](https://www.wired.com)

³ National Transportation Safety Board. Collision between a car operating with automated vehicle control systems and a tractor-semitrailer truck near williston, florida, may 7, 2016. Highway Accident Report NTSB/HAR-17/02, 2017. [ntsb.gov](https://www.ntsb.gov)

no human decided to stop. Years later a single trader was charged.

Required reading	<i>In der Strafkolonie</i> ⁴ by Franz Kafka (1919; Reclam UB 9900)
Preparation	Complete the Guided Reading Sheet individually before the session.

Guided Reading Sheet

Read *In der Strafkolonie* before the session. Work through the questions below individually, then bring your notes to the discussion.

1. The condemned man never learns what he is accused of or what the machine will inscribe on his body. What does this tell us about the relationship between legibility and legitimacy? Can a system be just if it cannot explain itself to those it acts upon?
2. Three figures surround the machine: the officer who understands and maintains it, the condemned man who cannot, and the visiting traveller who is invited to endorse it. How does responsibility shift across this triangle? Who, if anyone, is morally culpable for what happens?
3. The story illustrates the idea of the **responsibility gap**: when a complex system causes harm, no single human actor is clearly at fault. Can you name a contemporary case (in law, medicine, finance, or infrastructure) where this gap has appeared?
4. The EU AI Act (2024) requires human oversight for high-risk AI systems. What does "oversight" mean in practice if the system's decisions are too fast, too complex, or too numerous for humans to review individually?
5. Compare **causal responsibility** (who caused the harm) with **moral responsibility** (who is blameworthy) and **legal liability** (who must compensate). Do these three always converge? Give an example where they do not.

In-Class Experiment: The Responsibility Budget

A short allocation exercise to anchor the debate in shared experience. Run before the Opinion Landscape, using one concrete incident drawn from the cases above.

- **Setup.** State one incident in three sentences, for instance the hiring engine that filtered out a qualified applicant. On the board, list the actors: builder, deployer, operator, the people whose data trained the model, the regulator, the affected person, and no one.
- **Allocation.** Each student silently distributes one hundred responsibility points across those actors and writes a single sentence justifying their largest share. Collect the slips anonymously.
- **Tally.** Add the points on the board. The distribution will not converge: it spreads across five or six actors, and a visible share falls on no one. That spread is the responsibility gap, drawn from the room's own intuitions.
- **Debrief.** The two largest shares are almost always the provider (builder and deployer) and the user. Name them as the two sides of the debate to come, and have each student defend the allocation they made.

The EU AI Act's Risk Tiers

The EU AI Act⁵ is the first comprehensive attempt to close the responsibility gap by statute. It sorts every system into one of four tiers by the harm it can do, and attaches obligations that grow with the tier. The pyramid is narrow at the top: most systems carry no obligations.

- **Unacceptable risk, prohibited.** Social scoring by public authorities, subliminal manipulation, untargeted scraping of faces to build recognition databases, and most real-time biometric identification in public space. These are banned outright; no amount of documentation or oversight makes them lawful.
- **High risk, permitted but heavily regulated.** Systems that decide access to employment, credit, education, essential services, law enforcement, or migration. Before deployment each requires a risk management system, governed training data, technical documentation, event logging, human oversight, and a conformity assessment audited against the Act.
- **Limited risk, transparency only.** Chatbots, emotion recognition, and deepfakes. The single duty is disclosure: the user must be told they are dealing with a machine or with synthetic content.
- **Minimal risk, unregulated.** Spam filters, recommendation engines, game AI. This is the overwhelming majority of deployed systems, and the Act leaves them alone.

⁴ F. Kafka. *In der Strafkolonie*. Kurt Wolff Verlag, Leipzig, 1919. Reclam Universalbibliothek Nr. 9900

RISK-BASED REGULATION. Obligations scale with the stakes of the application, not the sophistication of the technology. A simple model deciding creditworthiness is regulated far more tightly than a complex one recommending films.

⁵ European Parliament and Council of the European Union. Regulation (eu) 2024/1689 laying down harmonised rules on artificial intelligence (ai act). Official Journal of the European Union, L series, 2024. Entered into force 1 August 2024

EVERY TIER IS A TRADE. Moving a system up the pyramid trades speed for protection. A minimal-risk feature can be deployed at once; a high-risk one waits on documentation, an audit, and a human-oversight design: months of work before deployment. The Act judges that where a system decides who gets a job, a loan, or bail, the price is worth paying. Logging lets a named party be held to account; human oversight keeps a person, not a dataset, the owner of the decision; the conformity assessment proves rigor before harm rather than after. Developers object: the same rigor slows European deployment against faster jurisdictions, and mandated transparency exposes model internals that are also security and competitive assets. Speed, user rights, security, and development rigor cannot all be maximised at once. The tiers fix the exchange rate between them, and the debate below turns on whether the Act has set that rate correctly.

Opinion Landscape and Debate

Format: Surrounded-style debate. Follow the shared debate protocol for the speaker's list rules and moderator scripts.

Opening question: When an AI system causes harm, who takes responsibility: the builder and deployer who put it into the world, or the user who chose to apply it to a purpose?

OPINION A, USER ACCOUNTABILITY. Responsibility rests with the user who put the system to use. Builder and deployer supply a capability; the user aims it at a purpose, and that choice carries the responsibility.

GENERAL-PURPOSE MODELS. Foundation models sit on a separate track: all owe transparency duties, and those trained above a compute threshold carry systemic-risk obligations whatever the downstream application.

The Builder insists the system performs within documented parameters and that responsibility follows its use.

- *Our documentation states that human review is mandatory before any recommendation is acted upon. The user relied on the output without it. We cannot be held liable for how our tool was used.*
- *A tool has no purpose until someone picks it up and gives it one. Using a tool for a purpose is an action, and the person who chose the purpose owns the outcome. We do not blame the scalpel for the surgeon's cut.*
- *Extend our liability to every downstream use and no one will build decision-support tools at all.*

The Deployer fielded the system and argues the final call sat with the user.

- *We procured a system certified for exactly this task, and it performed within specification. Our role is to make the capability available, not to make each individual decision with it.*
- *The final decision sat with the user. They read the output and chose to act on it. That choice, not our procurement, is the proximate cause of the harm.*
- *The user is a trained professional exercising discretion, and discretion carries responsibility. We cannot stand behind every judgement each operator makes.*

OPINION B, PROVIDER ACCOUNTABILITY. Responsibility rests with those who built and fielded the system. An AI is not a neutral instrument the user fully controls; it is closer to something its makers raised, and the user cannot answer for a decision they were never in a position to understand.

The User refuses to answer for a black box they could neither open nor overrule.

- *You handed me a black box and told me to trust it. I cannot open it, cannot audit the weights, cannot see why it decided what it decided. You cannot make a decision unauditably and then blame me for it.*
- *You call it a tool I chose to use, but a hammer does not learn. You trained this on data you chose and tuned it toward behaviour you selected. That is closer to raising a child than forging an instrument, and a parent answers for the child they shaped.*
- *The knowledge and the control both sat with the people who built and fielded it. Responsibility should sit where they do, not with the last human in the chain.*

The Regulator argues that whoever profits from and controls the design must carry the risk.

- *Whoever captures the upside must internalise the downside. A disclaimer buried in a manual cannot transfer a hazard the user has no means to evaluate.*
- *Product-safety law already binds those who build and field a product to its foreseeable real-world use, not to the idealised use their documentation describes.*
- *Placing the whole burden on the last human to touch the system builds a moral crumple zone⁶: a person positioned to absorb the blame while powerless to prevent the harm. That is not accountability; it is its appearance.*

INSTRUMENT OR AGENT? The law treats the two differently: you answer for a tool you wield; a maker answers for what it sends into the world. An AI sits between the two, and the debate turns on which side it falls.

KEY LEARNINGS The responsibility gap, the structural absence of a single culpable actor in distributed sociotechnical systems, is not caused by negligence but by the architecture of those systems: standard categories of fault (criminal intent, professional negligence, product liability) apply only awkwardly when causation is diffuse. Closing it requires deliberate governance design, not post-hoc blame. In practice the gap becomes a contest over allocation: builder and deployer point to documented limits and the user's discretion; the user points to an opaque product they could neither audit nor overrule. Beneath it lies one unsettled question: is an AI an instrument, whose user answers for its use, or a trained agent, whose makers answer for its formation? Explainability, the property of being able to account for outputs in terms a human decision-maker can evaluate, is a necessary but not sufficient condition for accountability; a system can be interpretable and still have no human willing to own its outputs. Meaningful human control, a regulatory concept that requires humans to retain the ability to understand, intervene in, and reverse AI decisions, is contested in practice: it may mean the ability to intervene, to understand, to reverse, or merely to authorise, and these are different requirements with different institutional costs. The lessons of aviation safety culture, medical liability law, and financial auditing all offer partial models for algorithmic governance, but none transfers cleanly.

Teaser: what happens to accountability when the system decides faster than anyone can follow or override?